Demo: Emulating Space Computing Networks with Rhone

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ABSTRACT

The rapid advancement in satellite technology with the adoption of commercial off-the-shelf (COTS) devices and satellite constellation networking has given rise to Space Computing Networks (SCNs). While SCN research is typically conducted on experimental platforms due to high operational costs, the unique challenges of SCNs require special consideration. In this demo, we introduce Rhone, an emulator that bridges these gaps by achieving both satellite- and constellation-level fidelity (the accurate replication of satellite and constellation states, including power, thermal, and network conditions, as well as application performance) while ensuring usability. Rhone adopts a two-phase approach: *i)* an offline phase builds power, thermal, orbit, network, and computation models using real satellite telemetries and hardware-in-the-loop chip mirroring, and *ii)* an online phase executes container-based emulation integrated with these models. Evaluation shows Rhone 's power and computation model errors under 5% and thermal model errors within 1.3 − 2.5°C.

CCS CONCEPTS

• **Networks** → **Network simulations**.

KEYWORDS

Space Computing Networks, Network Emulation

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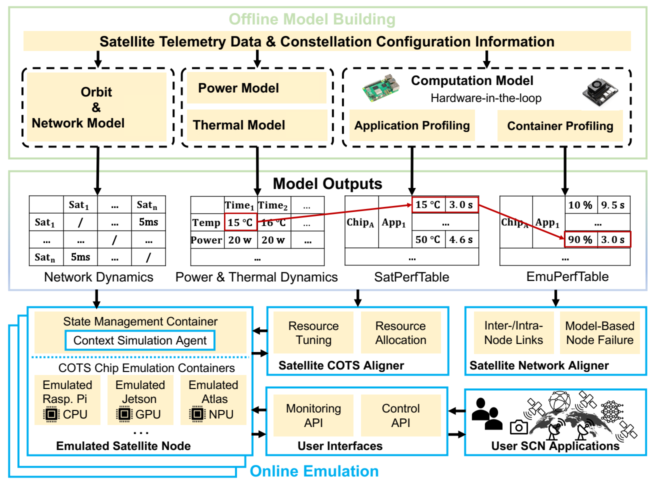

Figure 1: Rhone system overview.

1 THE RISE OF SPACE COMPUTING NETWORKS

Low-earth orbit (LEO) satellites are evolving from large, singular, and expensive nodes focused solely on communication into networked computing systems in space, which we term *Space Computing Networks (SCNs)*. However, SCN applications are hard to design, build and evaluate. We identify two unique characteristics of SCN that distinguish it from ground-based computing and networking: *i)* satellite COTS chips, which are not originally designed for space environment, face energy and thermal constraints. Overlooking these results in low application performance or even serious harm to the satellite such as overheat or power failure [1–4]; *ii)* the dynamic nature of constellation networks (*e.g.,* frequent topology changes [5–7] and node failures [1, 4, 8, 9]) leads to poor network stability and performance, and should be properly addressed.

2 MOTIVATION

Therefore, SCN applications must be designed, built and evaluated with careful consideration of the space environment to ensure proper operation and high performance. However,

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the high costs of operating or accessing a real constellation (*e.g.,* \$150,000 to launch a small satellite [10]) and increasing constellation sizes make it impractical to use real satellites for SCN research. Consequently, SCN research relies on practical experimental platforms that accurately replicate the space environment, *i.e.,* achieving *i) satellite-level fidelity*, involving heterogeneous COTS devices onboard and environmental context (*i.e.,* energy and thermal) they face; *ii) constellation-level fidelity*, involving complexity and dynamics of constellation networks, in terms of both topology changes and node failures; *iii) high usability*, scaling to encompass entire constellations while being cost-effective, flexible, and user-friendly. However, existing tools fall short of meeting all these requirements: *i) Physical setups* [11, 12] offer high fidelity but are difficult to access and lack scalability; *ii) Simulators* [13–18] use analytical modeling to simulate the space environment but lack system/network stacks and support for real system functionalities or interactive network traffic [19]; *iii) Emulators* [19–21] incorporate constellation network models with virtualized environments (*e.g.,* containers or virtual machines). However, they overlook heterogeneous satellite COTS computing devices facing space environment, especially energy and thermal limitations, and therefore cannot mimic COTS chips' performance, as our measurement study with real in-space satellites shows.

3 RHONE OVERVIEW

In this demo, we present Rhone, an SCN emulator that meets all the requirements above to support SCN research. The challenges in building Rhone stem from a conflict: As the computation performance of onboard chips is influenced by various factors like operating environment, hardware capacity, and application workload, it requires large system overhead (*e.g.,* CPU and memory) or high cost (*e.g.,* hardware device expenses) to achieve accurate emulation for individual satellites. However, while scaling to a constellation level, the total system overhead or cost for emulating SCN constellations exceeds the requirements for scalability and affordability. To address this, Rhone (Fig. 1) adopts a *two-phased* emulation process. *i) Offline model building phase*: we analyze telemetry data from in-space satellites and onboard COTS chips using *(a)* an open-source dataset [1] and *(b)* processed flight data which we collected from the Tiansuan constellation. We build power, thermal, orbit, and network models based on such data to replicate the space environment, and the computation model of COTS chips using *hardware-in-the-loop chip mirroring*; *ii) Online emulation phase*: we use container-based emulation integrating the models built in the first phase to mirror space environment. We also provide a set of easy-to-use monitor and command APIs for users to run unmodified applications.

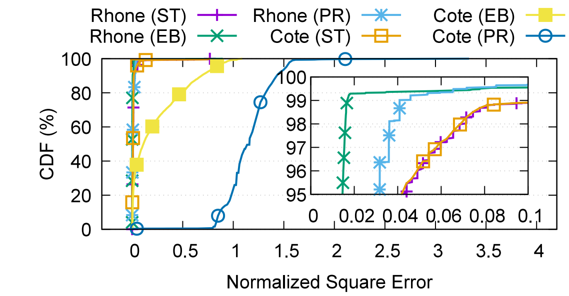

Figure 2: Energy harvesting accuracy.

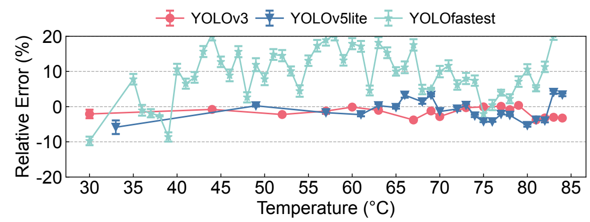

Figure 3: Computation emulation accuracy.

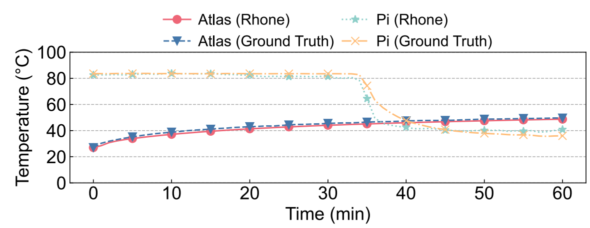

Figure 4: Temperature emulation accuracy.

4 EMULATION EVALUATION

We evaluate the scalability and fidelity of Rhone. Rhone shows $< 5\%$ error rate in power (Fig. 2) and computation models (Fig. 3), and $1.3 - 2.5°C$ average error in its temperature model (Fig. 4), compared to telemetry data we collect from in-orbit satellites as ground truth.

5 DEMONSTRATION SETUP

Our demonstration showcases Rhone 's emulation workflow and interactive capabilities. The setup is as follows:

- **Presenter hardware:** We will bring our own laptop to run the user-side terminal and visualization tools.
- **Connectivity:** We require a stable Wi-Fi or Ethernet connection to communicate with the emulation backend deployed on our remote server.
- **Demonstration workflow:** The laptop connects to the backend, submits SCN applications, monitors satellite- and constellation-level metrics in real time, and performs interactive control via Rhone 's APIs.

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